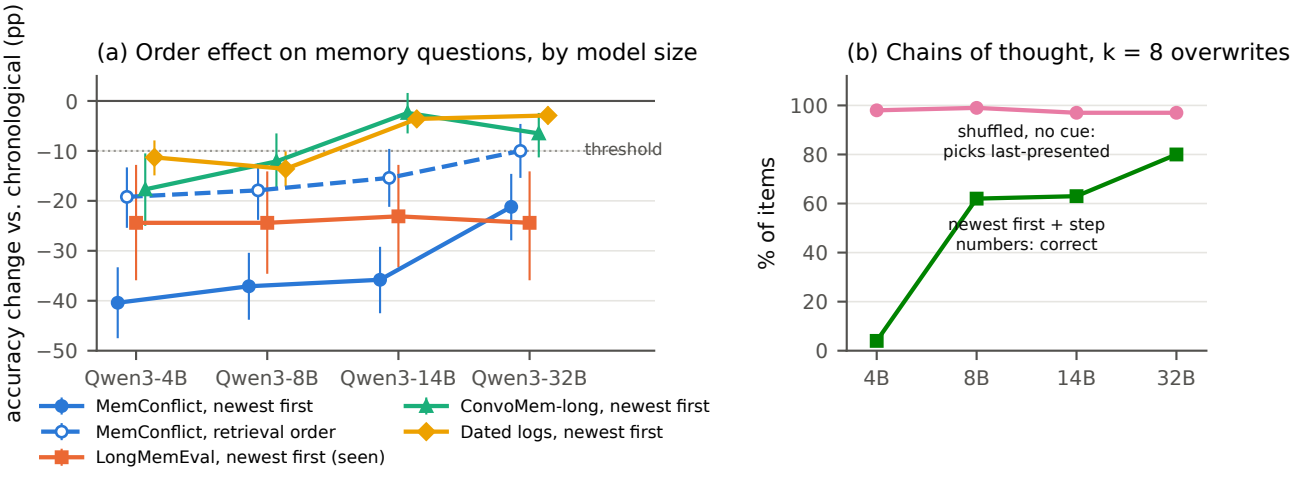
(a) Order effect on memory questions, by model size
accuracy change vs. chronological (pp)
0
−10
−20
−30
−40
−50
threshold
Qwen3-4B
Qwen3-8B
Qwen3-14B
Qwen3-32B
MemConflict, newest first
MemConflict, retrieval order
LongMemEval, newest first (seen)
ConvoMem-long, newest first
Dated logs, newest first
(b) Chains of thought, k = 8 overwrites
% of items
100
80
60
40
20
0
shuffled, no cue:
picks last-presented
newest first + step
numbers: correct
4B
8B
14B
32B